

Education

- 2023-27 **B.A.** Computer Science, **B.A.** Data Science^{*}, **Minor** Bioengineering, *University of California, Berkeley*
- 2020-23 **IB Diploma** Higher level Math AA, Biology, Chemistry, *International English Gymnasium Sodermalm*

Research Experience

- 2025– ML Researcher, [Yun S. Song Lab](#), [Innovative Genomics Institute](#) & Berkeley Artificial Intelligence Research. (Advisors: Yun S. Song, Rohan Sachdeva, Spencer Diamond)
- 2023– Computational Immunology Researcher, Borch Lab, Washington University in St. Louis. (Advisor: [Nicholas Borcharding](#))
- 2024 Bioinformatics & AI engineer, [Generation Lab, Inc.](#) (YC S24)
- 2023-25 Researcher & Bioinformatician, Irina M. Conboy Lab, University of California, Berkeley & [QB3](#).
- 2023 Researcher, Hallucinating Scaffolds team, [iGEM at Berkeley](#). (Advisor: [Chris Anderson](#))
- 2023 NGS Method developer, [Sahlin Group](#), SciLifeLab & Stockholm University. (Advisor: [Kristoffer Sahlin](#))
- 2022-23 Researcher, [Karlsson-Hedestam/Murrell Lab](#), Karolinska Institute. (Advisor: [Benjamin Murrell](#))

Awards & Honors

- 2025 Dean's List - UC Berkeley College of Statistics, Data Science, and Society
- 2023 Finalist, National Swedish Research fair ([1/55](#))
- 2023 Top 60, National Swedish Programming Olympiad Site
- 2022 Top 115, National Swedish Chemistry Olympiad
- 2022 Double Admit to Karolinska Institute's summer research programs (1/5 admitted nationwide)
- 2022 Top school participant, Swedish Biology Olympiad

^{*}With [Domain Emphasis](#) of Computational Methods in Molecular and Genomic Biology

Fellowships & Funding

- 2025 Top 20 Global Fellow, competitive BioTech/AI entrepreneurship program; eligible for mid six-figure funding (< 0.1% acceptance rate)
- 2025 Recipient, Anna Whitlock Scholarship, [Anna Whitlock Memorial Foundation \[In Swedish\]](#)
- 2023 Recipient, PostHS scholarship, [Equitable College Counselling](#)

Academic Service

- 2026 Abstract Reviewer, Gerontological Society of America (GSA) Annual Scientific Meeting

Publications

† → Equal contribution

Journal Articles

- J1. Cruz, J. M., Yeung, H., Alzalzalee, R., **Yang, Qile**, Kabir, H., McDonough, S., Mei, X., Conboy, M. J. & Conboy, I. M. In old mice, Exercise Induces Inflammation and Fibrosis unless Alk5-inhibitor and Oxytocin are used. *Journal of Cellular Physiology* **240**, e70054. <https://onlinelibrary.wiley.com/doi/abs/10.1002/jcp.70054> (June 2025).
- J2. **Yang, Qile**, Safina, K. R., Nguyen, K. D. Q., Tuong, Z. K. & Borcharding, N. scRepertoire 2: Enhanced and efficient toolkit for single-cell immune profiling. *PLOS Computational Biology* **21**, 1–10. <https://doi.org/10.1371/journal.pcbi.1012760> (June 2025).
– Media: [ASCA](#), [Chinese Software Developer Network \[In Chinese\]](#), [Jianshu \[In Chinese\]](#)
– BioRxiv preprint: <https://doi.org/10.1101/2024.12.31.630854>.
- J3. **Yang Qile**. APackOfTheClones: Visualization of clonal expansion with circle packing. *Journal of Open Source Software* **9**, 6868. <https://doi.org/10.21105/joss.06868> (Nov. 2024).
– Media: [R views \(by Rstudio\)](#), [R-bloggers](#), [Zhihu \[In Chinese\]](#), [360doc \[In Chinese\]](#), [Tencent \[In Chinese\]](#).

Working Papers

- W1. Cruz, J. M., Alzalzalee, R., Yeung, H., Mahmood, Z., **Yang, Qile**, Morshedian, N., Robinson, Z. R., Malhotra, K. R., Conboy, M. J., Mazahery, A. R., Nevado, J. B. & Conboy, I. M. *Plasma dilution after myocardial ischemia-reperfusion injury promotes cardiac repair, heart performance, and recovery of motor function and endurance in old mice* Manuscript In Press. 2025.
- W2. Borcharding, N., **Yang, Qile**, Sun, B., DeNardo, D., Mudd, P. A., Ellebedy, A. & Brestoff, J. R. *Ibex: Deep autoencoders for single-cell BCR sequencing* Manuscript under preparation. 2025. <https://github.com/BorchLab/Ibex>.

Presentations

Posters

- P1. **Yang, Qile**, Sahlin, K. & Murrell, B. *KmerGMA: A seed-based approach for fast homology searching* Swedish National Science Fair Finals (Södertälje, Sweden). Apr. 2023. <https://qile0317.github.io/KmerGMA.jl/stable/>.
- P2. **Yang, Qile** & Borcharding, N. *Reimagining Single Cell BCR Modeling: An Interpretable, Biochemically-Informed Framework* Cal Undergraduate Bioengineering Symposium. Dec. 2025. <https://igem.studentorg.berkeley.edu/cubs/>.
- P3. **Yang, Qile** & Borcharding, N. *Why use a pLM when Tiny Autoencoders will do? Dissecting Antibody Representations and Scaling to Single-Cell BCRs with Ibex* Bay Area Bioengineering Symposium. Apr. 2026. <https://igem.studentorg.berkeley.edu>.

Demonstrations & Tutorials

- D1. **Yang, Qile** & Murrell, B. *Discovery of Novel Camelid Germline Immunoglobulin Alleles* Karolinska Institute's Summer Research Program in Computational Biology (Stockholm, Sweden). Aug. 2022. <https://github.com/Qile0317/SoFoCompBio22>.
- D2. **Yang, Qile et al.** *Final Presentation: Hallucinating Scaffolds Team* iGEM at Berkeley All Hands 3 (Berkeley CA, USA). Dec. 2023.

Software

- S1. **Yang, Qile**. *Einops for R: Flexible, Powerful, and Readable Tensor Operations* The Comprehensive R Archive Network. 2025. <https://qile0317.github.io/einops/>.
- S2. **Yang, Qile**. *FastUtils: Fast, Readable Utility Functions* The Comprehensive R Archive Network. 2025. <https://qile0317.github.io/FastUtils/>.
- S3. Lin⁺, X., Lo⁺, K. & **Yang⁺, Qile**. *OKaiLoRa: a no-code platform for training, running, and sharing machine learning models in healthcare* Devpost. 2024. <https://devpost.com/software/okailora-ai>.

Selected Media Coverage

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| 2025 | Association of Single Cell Analysis (ASCA),  scRepertoire 2: Enhanced and efficient toolkit for single-cell immune profiling [J2] |
| 2024 | Chinese Software Developer Network, Immune repertoire analysis - scRepertoire [in Chinese] [J2] |
| 2023 | R-views (by RStudio), May 2023: "Top 40" New CRAN Packages [J3] |

Selected Professional Organizations

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| 2026– | External Relations Team Member, iGEM @ Berkeley |
| 2025– | Operations & Tech Lead, Computational Biology @ Berkeley |

2023– Member, [Bioconductor Developer Community](#)
 2025– Member, [Gerontological Society of America \(GSA\)](#)
 2025– Member, [The Antibody Society](#)
 2025– Member, [Boston Protein Design and Modelling Club \(BPDMC\)](#)

Other Distinctions

2023 Swedish National Debate Team member, [Swedish Schools Debating Organization](#)
 2023 World Schools Debating Championship Sponsorship (Withdrew due to external constraints), [The Scholarship Fund in Memory of Ellinor Carlsson \[In Swedish\]](#)

Relevant Coursework

* → Taking currently

Computational Biology: Probabilistic Modeling in Computational Biology* | Computational Single-Cell and Systems Immunology | Machine Learning, Statistical Models, and Optimization for Molecular Problems | Computational Functional Genomics | Algorithms for Computational Biology* | Notable Discoveries in Systems and Computational Biology over the last 50 years*

Machine Learning / Statistics / Maths: Introduction to Machine Learning | Introduction to Artificial Intelligence | Concepts of Probability | Linear Algebra and Differential Equations | Multi-variable Calculus | Discrete Mathematics and Probability Theory

Biology / Chemistry: General Genetics* | Frontiers in Microbial Systems Biology* | Chemical Structure and Reactivity | Organic Chemistry Laboratory | Introduction to Bioengineering | Ethics in Science and Engineering

Computer Science: Structure and Interpretation of Computer Programs | Data Structures | Introduction to Competitive Programming

Last updated: April 26, 2026